

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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The basic structure of FIR digital filter

- Direct Form Structure
- Linear Phase Direct Form Structure
- Cascade Structure

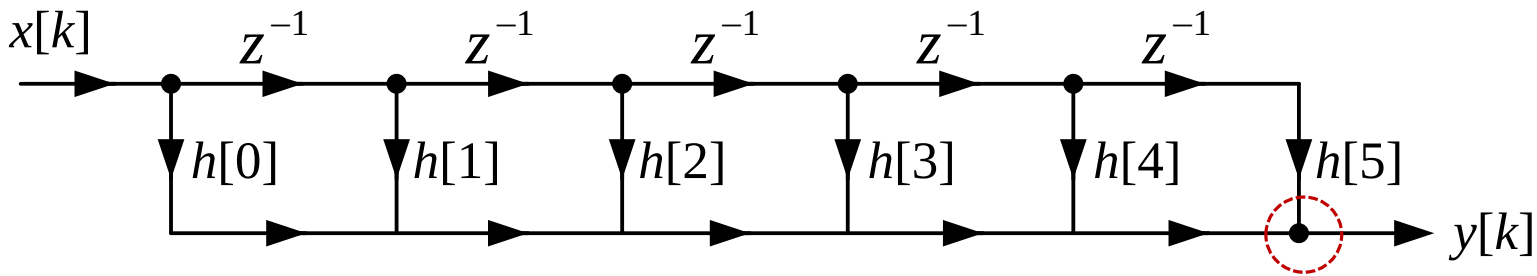
The basic structure of FIR digital filter

The system function of an *M*-order FIR digital filter is:

$$H(z)=\sum_{k=0}^{M} h[k]z^{-k}=\sum_{i=0}^{M} b_i z^{-i}$$

Let the order of the FIR digital filter be M = 5

$$H(z)=h[0]+h[1]z^{-1}+h[2]z^{-2}+h[3]z^{-3}+h[4]z^{-4}+h[5]z^{-5}$$



The direct form structure of a linear-phase FIR digital filter

When an FIR digital filter has linear phase, its impulse response $h[k]$ exhibits symmetric characteristics:

$$h[k] = \pm h[M-k]$$

When implementing the direct form structure of a linear-phase FIR digital filter, multipliers can be shared to reduce costs, resulting in the direct form structure of a linear-phase FIR digital filter

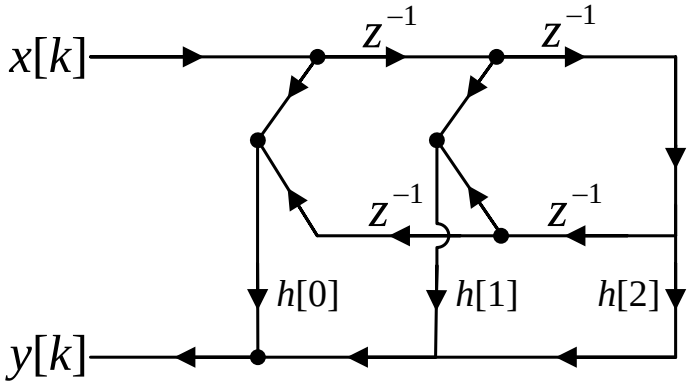
The direct form structure of a linear-phase FIR digital filter

Type I FIR Filter System: $M=4$, $h[k]=h[M-k]$

$h[0]=h[4]$
$h[1]=h[3]$

$$H(z)=h[0]+h[1]z^{-1}+h[2]z^{-2}+h[1]z^{-3}+h[0]z^{-4}$$

$$=h[0](1+z^{-4})+h[1](z^{-1}+z^{-3})+h[2]z^{-2}$$



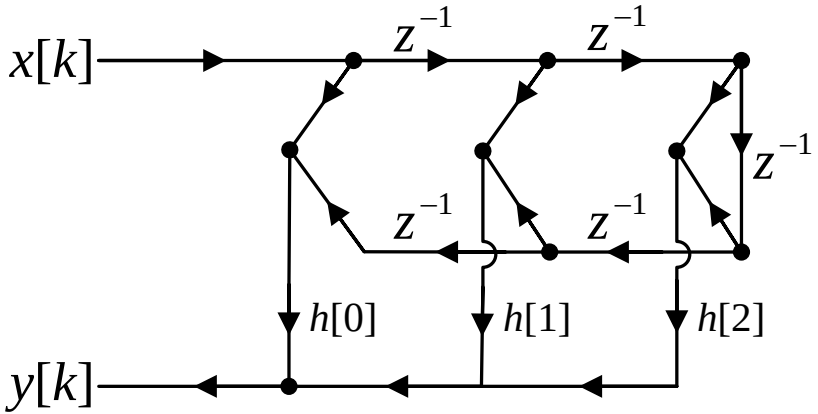
The direct form structure of a linear-phase FIR digital filter

Type II FIR Filter System: $M=5$， $h[k]= h[M-k]$

$h[0]=h[5]$
$h[1]=h[4]$
$h[2]=h[3]$

$$H(z) = h[0] + h[1]z^{-1} + h[2]z^{-2} + h[2]z^{-3} + h[1]z^{-4} + h[0]z^{-5}$$

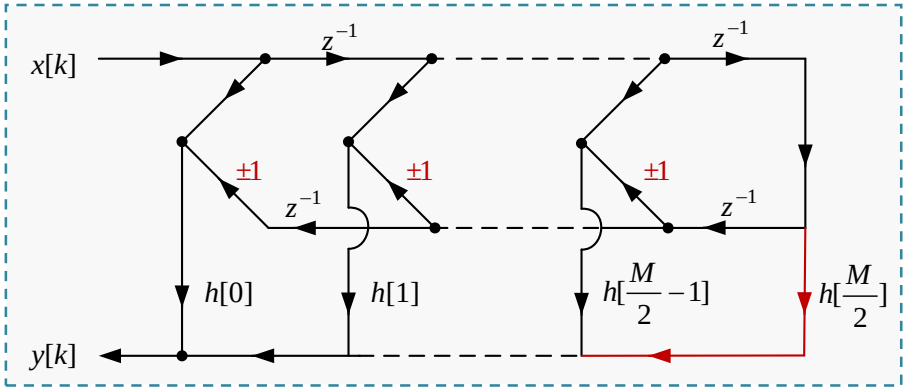
$$= h[0](1+z^{-5}) + h[1](z^{-1}+z^{-4}) + h[2](z^{-2}+z^{-3})$$



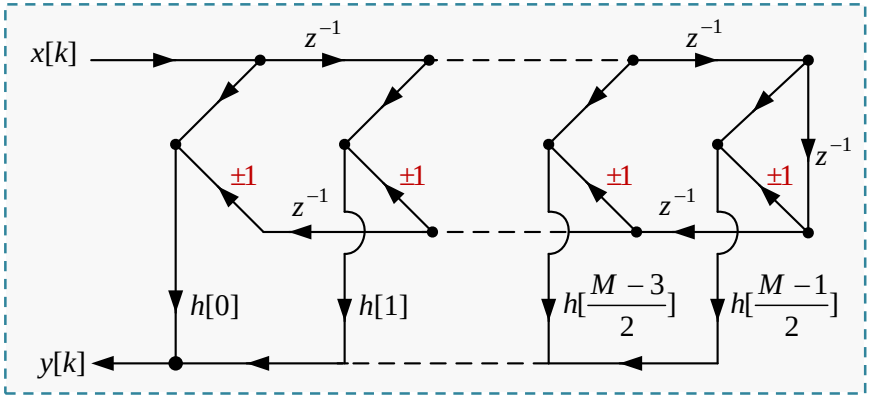
The direct form structure of a linear-phase FIR digital filter

The four types : (I, II, III, IV) of linear-phase FIR digital filters

$$h[k] = \pm h[M-k]$$



M is even (Type I, Type III)

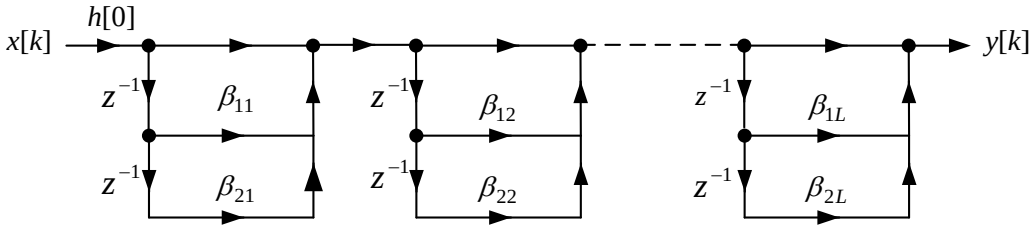


M is odd (Type II, Type IV)

The cascade structure of FIR digital filter

If $H(z)$ is factored into several first-order or second-order factors with real

$$H(z) = h[0]\prod_{k=1}^{L}(1+\beta_{1,k}z^{-1}+\beta_{2,k}z^{-2})$$



A cascade structure is obtained, allowing individual control of the zeros of each subsystem.

Examples of FIR digital filter structures

[Example] Given the system function of a certain FIR digital filter as

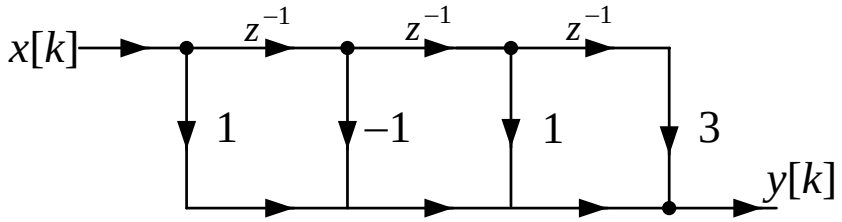
$$H(z) = (1 + z^{-1})(1 - 2z^{-1} + 3z^{-2})$$

Try to draw both the direct form and cascade form structures.

Solution: Direct Form

Express the system function $H(z)$ as:

$$H(z) = 1 - z^{-1} + z^{-2} + 3z^{-3}$$



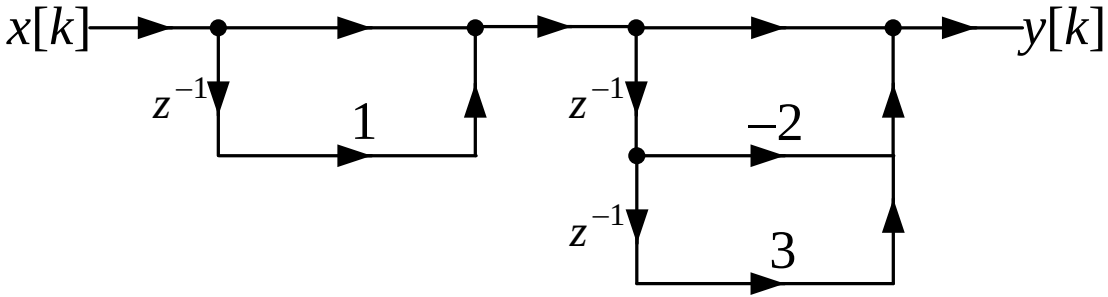
Examples of FIR digital filter structures

[Example] Given the system function of a certain FIR digital filter as

$$H(z) = (1 + z^{-1})(1 - 2z^{-1} + 3z^{-2})$$

Try to draw both the direct form and cascade form structures.

Solution: Cascade Form

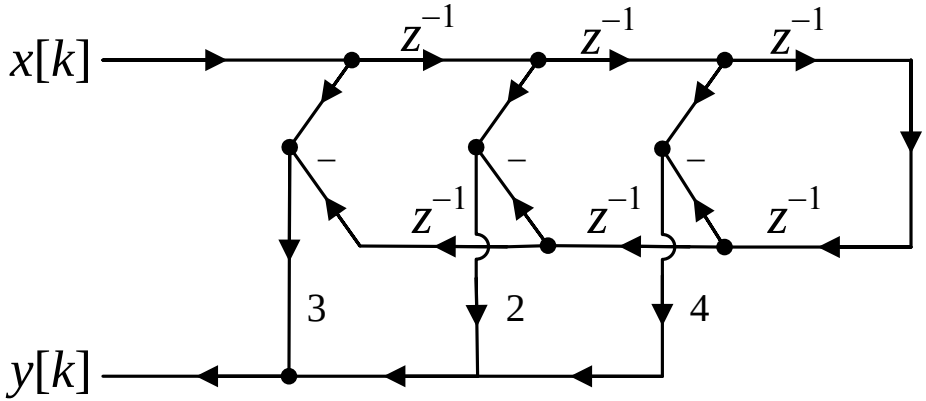


Examples of FIR digital filter structures

[Example] Given the system function of a 6th-order linear-phase FIR digital filter (Type III) with the unit impulse response $h[k]=\{3, 2, 4, 0, -4, -2, -3\}$, which satisfies $h[k]=-h[M-k]$, draw the linear-phase direct form structure of this FIR digital filter.

solution:

$h[0]=-h[6]=3,$
$h[1]=-h[5]=2,$
$h[2]=-h[4]=4,$
$h[3]=0$

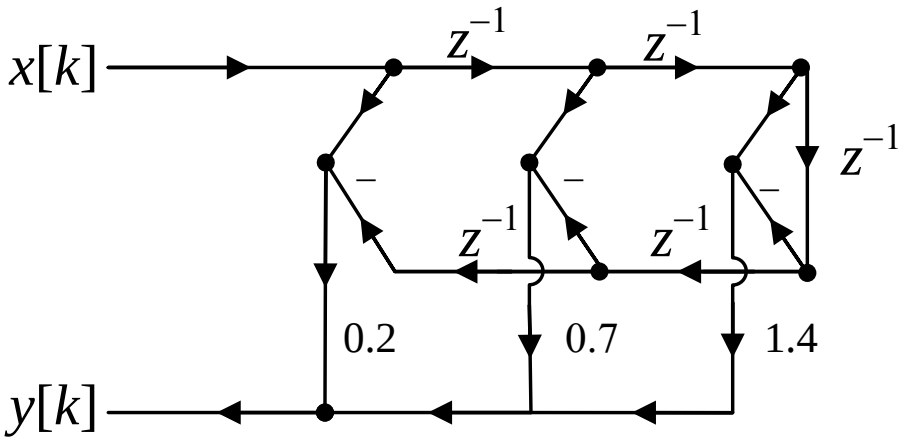


Examples of FIR digital filter structures

[Example] Given a 5th-order linear-phase FIR digital filter (Type IV) with the unit impulse response $h[k]=\{0.2, 0.7, 1.4, -1.4, -0.7, -0.2\}$, which satisfies $h[k]=-h[M-k]$, draw the linear-phase direct form structure of this FIR digital filter.

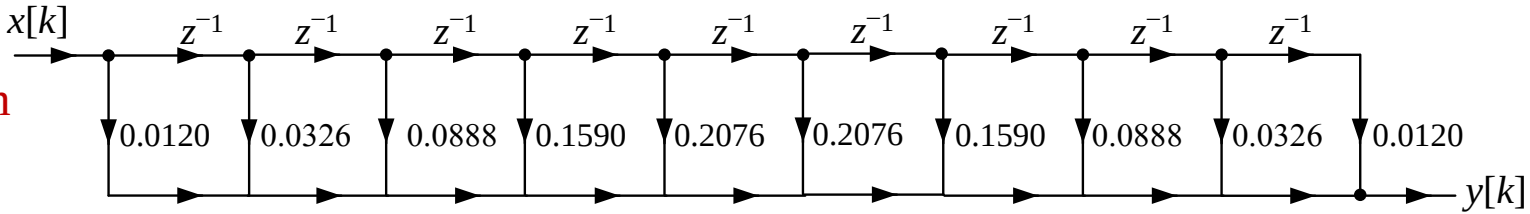
solution:

$h[0]=-h[5]=0.2,$
$h[1]=-h[4]=0.7,$
$h[2]=-h[3]=1.4,$

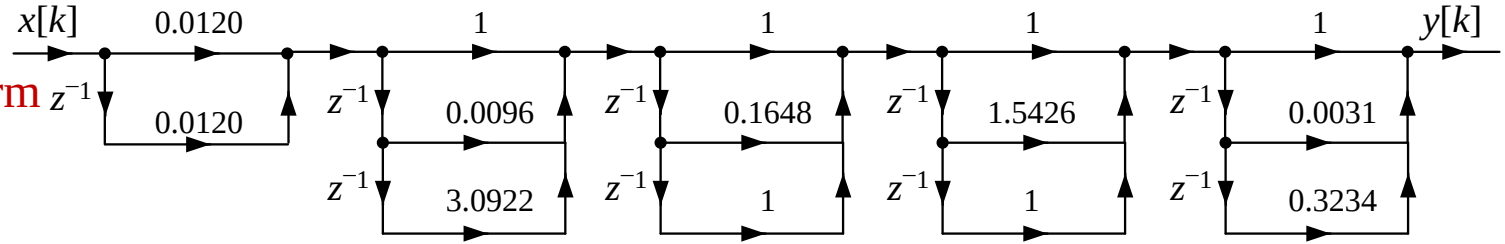


Use MATLAB to implement a digital filter

Direct Form



Cascade Form



Use MATLAB to implement a digital filter

[Example] Use MATLAB to design a 9th-order linear-phase FIR digital low-pass filter with a cutoff frequency of 0.1π rad,Find the coefficients for the cascade structure of this filter.

```
M = 9;
h = fir1(M,0.1,hamming(M+1));
sos = tf2sos(h,1);
[row col] = size(sos);
for i = 1:row
    for j = 1: col
        fprintf('%2.4f\t', sos(i, j));
    end
    fprintf('\n');
end
```

Direct Form FIR digital filter coefficients：

0.0120	0.0326	0.0888	0.1590	0.2076
0.2076	0.1590	0.0888	0.0326	0.0120

Cascade form FIR digital filter coefficients：

0.0120	0.0120	0.0000	1.0000	0.0000	0.0000
1.0000	0.0096	3.0922	1.0000	0.0000	0.0000
1.0000	0.1648	1.0000	1.0000	0.0000	0.0000
1.0000	1.5426	1.0000	1.0000	0.0000	0.0000
1.0000	0.0031	0.3234	1.0000	0.0000	0.0000

Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!